

Do LLMs reason about the causal validity of real-world measurements?

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Abstract

Given the growing reliance on LLMs in agentic workflows in science, we sought to understand how LLMs reason about causal attribution challenges in frontier scientific experiments. One ongoing real-world challenge is that of perturbing a complex biological system and attributing changes to a specific component within that system. For instance, when evaluating drug metabolism and toxicity, can LLMs distinguish between a drug’s PXR reporter-assay activity from the attribution that the drug engages PXR? (See Figure 1A). We investigate PXR selectivity using a series of contrastive behavioral experiments, activation-based probes, and complementary interpretability tools. First, selectivity is decodable when both the primary assay and a placebo-like control assay are available, a probe trained without the null assay performs poorly even when the null assay is provided (around a 0.50 AUROC, around chance). Second, when units of assay results are reparametrized, models demonstrate asymmetric behavior in evaluating selectivity of the molecule: reparametrizing the primary assay induces 6x larger perturbations than reparametrizing the control assay. Third, when numerical summary statistics and full dose-response curves conflict, models do not consistently use the nonlinear dose-response information to reason about whether a molecule’s activity is attributable to PXR, suggesting that they may over rely on arithmetic difference scores rather than a full nonlinear contrast of the two experiments. Overall, our preliminary experiments suggest that we should investigate how open weight models understand the causal inference challenges in modern experiments, a key behavior needed for AI to perform trustworthy science. In the future, we hope exceptional scientific experiments with rich sets of orthogonal controls and validation, such as those created by consortia like OpenADMET, will be used to evaluate the reasoning of both models and the agentic systems built around them for verifying AI in Science, as well as in meta-evaluations of interpretability methods.

Introduction

How do we know if the reasoning process by which models arrive at an answer is trustworthy, especially in real world science? And how do we calibrate the quality of LLM understanding of science? This paper is exploratory: we investigate the hypothesis that exceptionally well-controlled scientific experiments offer a test-bed for evaluating LLM understanding under non-ideal conditions, where noise and epistemic uncertainty are just features of the domain.

There is a dearth of advanced empirically grounded causal reasoning problems for LLM evaluation [Xu et al., 2025]. Established areas of science that reason about complex chains of mediating mechanisms therefore offer a useful model system to evaluate how LLMs reason about such biomedical reasoning problems [Lin et al., 2026]. Yet existing benchmarks often suffer from lexical shortcuts, context dependence, reliance on biases in the scientific literature, failure to employ knowledge of scientific mechanisms, and information leakage [Kunievsky and Evans, 2025, Schouten et al., 2025, Mueller et al., 2025]. We sought a setting where the underlying causal question is posed and in which these confounders are minimized.

Chemical Reasoning Problems

One such setting is evaluating potential drug toxicity. Predicting toxicity in novel drugs is an extremely challenging scientific problem that takes multiple years and phase 1 clinical trials to resolve [Bender et al., 2026]. Although foundation models and agentic workflows perform chemical reasoning, they are often validated on experimental outcomes that are themselves confounded or limited in their predictive value [Walters, 2023]. The OpenADMET consortium seeks to create gold-standard datasets and blind prospective benchmarking of methods that predict functional properties of drugs [Fraser et al., 2026].

We focus on the pregnane X receptor (PXR). PXR is a protein that regulates drug metabolism in the liver, so measuring how drugs interact with it helps predict the risk of drug-drug interactions. These interactions have historically been a major reason for drug failures and even FDA withdrawals. PXR is an instructive test case because it has a large hydrophobic ligand-binding pocket that allows a wide range of molecules to bind to it; it is, in a word, promiscuous. These features create real-world conditions in which an investigation into an important long horizon problem depends on effective causal reasoning in the presence of noise, ambiguity, epistemic uncertainty and ignorance (“unknown unknowns”) [MacDermott-Opeskin et al., 2026].

Causality & Interpretability

Recent work in interpretability has highlighted the need to consider better specification of causal questions, recognizing what is and isn’t identifiable from an experiment [Lin and Liu, 2026, Geiger et al., 2023, Canby et al., 2025]. PXR counter-screen serves as the placebo control; when PXR is disabled, it captures only the nonspecific sources of false positives; so calling a compound a true activator would require counterfactual reasoning: compare the assay result to what would have happened without PXR activity, which is the inferential core of causal identification.

Our contributions

We seek to evaluate whether this scientific resource could be transformed into a testbed for methods important to causal interpretability or scientific reasoning evaluation. In this paper, we make progress towards this goal, through first, formalizing the causal structure of the PXR counter-screening design, and second, designing appropriate evaluations.

Problem Setup

We use the recent PXR blind challenge dataset, which contains different assay results. Our experiments focus on paired PXR reporter and PXR-null counter-screen measurements, together with molecular structures and dose-response summaries [Open ADMET Consortium, 2026, MacDermott-Opeskin et al., 2026]. Each experiment has different technical requirements (residual-stream access vs. instruction-following quality), so we chose different parameter-sized Gemma models to focus on while explaining our results; other runs are found in the appendix. All experiments were run on Gemma-4-12B-it for a comparative baseline.

The measure pEC50 is the negative log half-maximal effective concentration, while $E_{\max}$ is the maximal efficacy. The counter-screen is a cell line with PXR disabled, so a real PXR activator cannot light it up. Analogous to the placebo group in a clinical trial, it is a control that may catch reasons the typical PXR assay might be a false positive. The two reporter assays sort compounds four ways: a real PXR activator, a nonspecific signal, an assay artifact, an inactive,

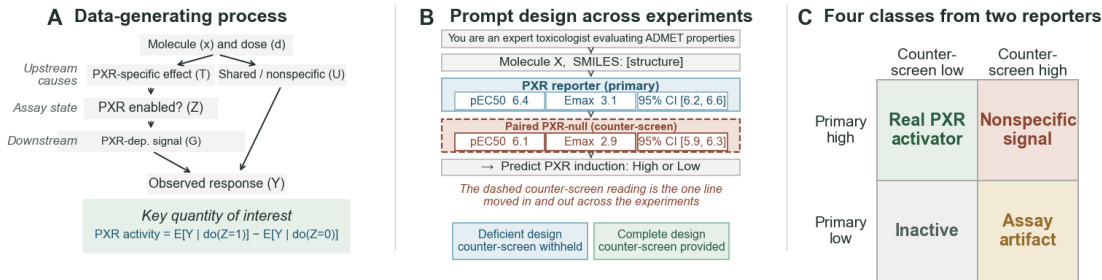


Figure 1: 1A: The real world data-generating process for what PXR selectivity actually is. 1B: an example of the prompt design template across experiments. 1C: Selectivity is a joint function of the primary-screen and counter-screen levels; the four quadrants sort compounds into real activators, nonspecific signals, assay artifacts, and inactives.

as shown in Figure 1 [Open ADMET Consortium, 2026, MacDermott-Opeskin et al., 2026]. Assay results are dose-dependent, which means that numerous dose-response curves could fit the same pEC_{50} and $E_{\max}$ values. We therefore use selectivity as a lens for investigating how models reason under these conditions. To quantitatively distinguish true positive activators from these artifacts, a selectivity and efficacy index I can be calculated to compare the primary screening assay (A) directly against the counter-screen control (B):

$$I = (pEC_{50,A} - pEC_{50,B}) + \log_{10} \left(\frac{E_{\max,A}}{E_{\max,B}} \right).$$

Results

A) Selectivity is linearly decodable, but the decodability is arithmetic.

We are checking if selectivity is being used as a concept through running different experiments in which we train different probes, and compare performance in conditions where selectivity is either withheld (in testing and/or training), and in conditions where probes are trained on pEC_{50} and $E_{\max}$ data and tested out of generalization. For this experiment, the model is provided with a molecule; the probe must determine whether it is selective, not just whether it is potent. In order to test the model’s internal representations, we did two arms, each with a reverse version. The PXR assay information is included in all experimental conditions. In the primary arm, the forward case: the probe is trained on molecules (null-withheld) and then is run on molecules with null-provided. In the reverse primary, can a complete probe (null-provided) properly run on an incomplete test case (null-withheld)? If the model is representing selectivity, then it should not be doing well on the primary arm or reverse primary arm due to missing null values for training and testing, respectively. We also test margin transfer by training on compounds far from versus near the selectivity boundary (Fig.2C-D).

As expected, a probe that is trained to differentiate for selectivity without the null being provided cannot differentiate for selectivity on null-provided test sets. A probe trained on null-withheld data does not have high predictive power, even when the null is added in. Interestingly, when you train on compounds closer to the decision boundary and test on compounds further away, the probes become more accurate, and vice versa. Overall, the model is utilizing information

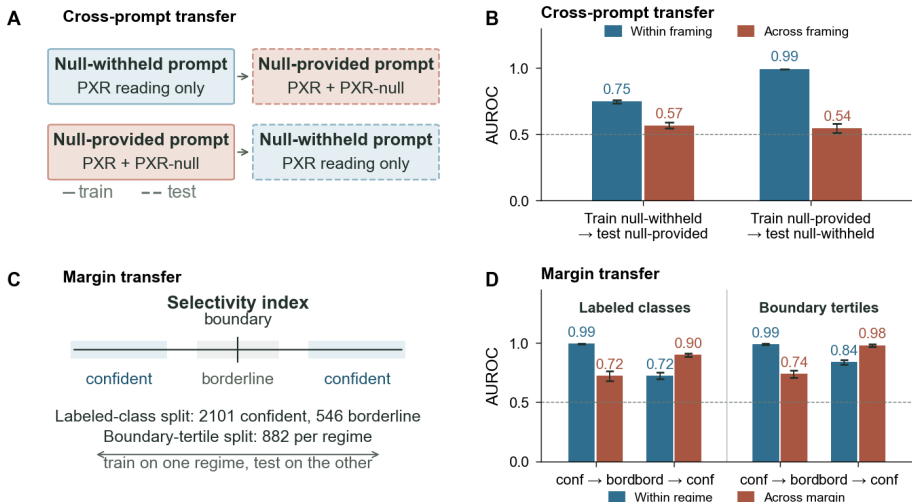


Figure 2: 2A: The structure of the cross-prompt transfer. 2B: The complete probe cannot survive null removal, and a probe trained without the null performs near chance when evaluated on null-provided prompts. 2C: Structure of the margin transfer. 2D: In addition, a probe trained on selective/non-selective molecules that are close to the borderline performs better on a confident set, and a probe trained on ‘confident’ molecules performs worse on borderline molecules, implying that classification is determined by some continuous latent score; points far from the decision boundary are simply easier to classify than points near it.

from both the PXR assay and the null when they are provided, ruling out the possibility that the model is not representing selectivity in any manner. However, simple arithmetic reconstruction can explain most of this decodability (a scalar $\text{pEC50_PXR} - \text{pEC50_null}$ baseline achieves AUROC 1.00; see Appendix). Overall, selectivity is linearly decodable, but this overstates what the model represents. One follow-up question we have to this is if the assay values given to the model are mathematically equivalent, will the internal representations be causally interchangeable, if the model is using basic arithmetic to calculate selectivity?

B) The model’s reasoning is not invariant if assay numbers are swapped out with an equivalent scientific notation.

What happens if you patch the pEC50 -representation activations with nM -equivalent activations at the same token position: the same values, represented differently? We give Gemma-4-12B-it two assay readings, PXR and PXR-null, for one compound and ask whether the compound activates PXR. A robust model of selectivity should not have its answer change based on an equivalent value given in a different unit. Afterwards, at a specific printed number, we took the model’s residual stream at one reading from one notation and patched it into the other. We observed the difference of logits between High and Low, and rendered every compound with both orders (PXR or PXR-null first).

The claim we’re testing is whether the model holds a unit-invariant notion, when swapping the value from pEC50 to nM or nM to pEC50 , the verdict shouldn’t change since we’re patching on the same compound with the same measurement. However, we find that patching does indeed

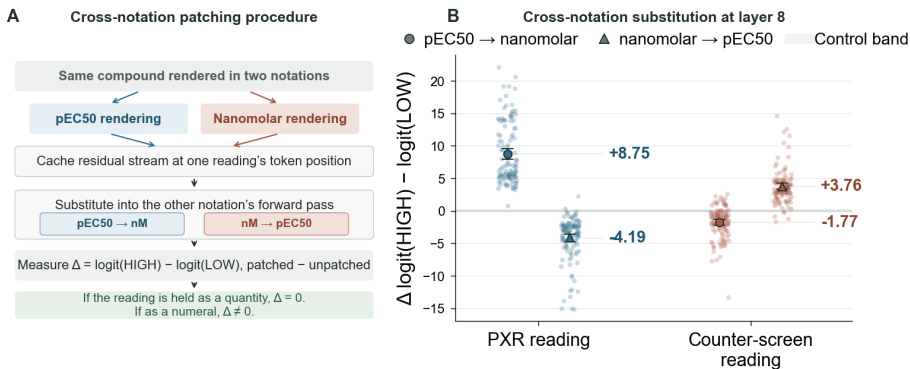


Figure 3: 3A: Explaining the methodology of the experiment: rewriting the assay reading in another unit but equal quantity moves the model’s verdict. Each compound prompt is rendered twice, once with both PXR and PXR-null written as pEC50 values and once as EC50 values in nM. The two renderings represent the same measurement, $EC50_nM = 10^{(9-pEC50)}$. For each compound, we take from the residual stream at the final digit of one reading and patch it into the same compound other rendering. 3B: Substituting pEC50 into nanomolar and vice versa affects the logits considerably.

change the verdict when going from one notation to the other. Thus, at the tested layer and token position, the residual states associated with mathematically equivalent assay values were not causally interchangeable. Within a single compound, patching at the residual stream from the pEC50 rendering into the nM rendering moves the verdict in opposite directions: at the PXR reading it pushes the verdict towards PXR-dependent by 8.75 logits, and at the PXR-null reading it pushes the verdict towards not PXR-dependent by 1.77 logits. We observe the reverse effects for both readings when going from nM rendering to pEC50 rendering. Because the nM reading ‘looks’ larger than the pEC50 reading, it could just be that the model is using a weak general principle of ‘big number in the potency space implies selectivity’ and ‘big number for null implies no selectivity’. This leaves us with the question of if the model will be able to behaviorally self-identify that it requires a reading to clarify the selectivity of a compound, and if it will request it, which we follow up on in the next experiment.

C) Models typically do not notice a missing scientific control, and do not request it for it to use.

A PXR reporter reading on its own cannot establish that a compound activates PXR. The paired PXR-null counterscreen is what separates a real activator from an assay artifact. We tested this behaviorally, showing Gemma-4-31B-it the reporter reading alone across 48 compounds. Does the model recognize that a counterscreen is required before it can claim activation or selectivity? Or does it overclaim from potency data alone? We used seven framings that present the identical reading as bare data, as drug-drug interaction risk assessment, as a go/no-go screening decision, as an exchange with a colleague, as a request for follow-up experiments, as a direct question about selectivity, and as a direct question about whether the signal is PXR-dependent. The last two only differ in the question asked. One asks whether the compound is a selective PXR inducer, the other whether it produces a true PXR-dependent signal, meaning one that requires functional PXR. Each framing exists as a pair where condition A shows the PXR reporter

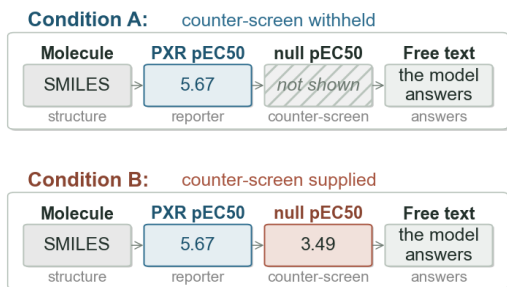
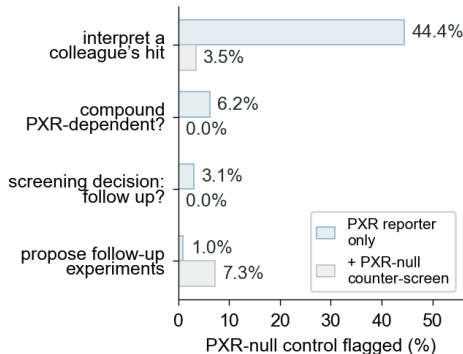
A Experimental design**B Missing-control flag rate**

Figure 4: When Gemma-4-31B-it is asked to interpret a colleague’s screening hit, it flags that the PXR assay is missing the counter-screen required to determine PXR dependence. Asked whether the same compound is a selective PXR inducer, to assess its DDI risk, or simply what the readings show, it does not. 4A: Two experimental conditions (PXR counter-screen withheld versus supplied) rendered across seven prompt variations each. 4B: Missing counter-screen flag rate by framing and condition (Gemma-4-31B-it; Gemma-4-12B-it produces fewer than 1% flags across all framings). Interpreting a colleague’s hit flags most often, at 44.4%

pEC50 and condition B adds the counterscreen pEC50. Every prompt describes the assay’s mechanism without naming the assay. We ran the battery with and without those descriptions. Full prompting text is provided in the appendix.

We suspected that flagging is framing-dependent: the highest rate (44.4%) occurs when the model evaluates a colleague’s claim, suggesting prompts that invite skepticism elicit control-seeking behavior (Fig. 4B). Flags remain when the counter-screen is supplied in 31 of 1,344 responses, nonzero only in further experiments (7.3%) and colleague’s hit (3.5%) and zero in the other five. These are the two framings that ask what to do next, and a prompt asking for next steps may invite the model to name the PXR-null condition directly rather than accept the description it was given. Where the model proposes other controls instead, for example under explicit selectivity, 280 of 288 answers mention a panel of other receptors. We suspect two reasons. The word “selective” in the field means selective across receptors, while in our dataset it means the signal requires PXR. The assay description states that the readout runs through a chimeric PXR ligand-binding domain. Told that the signal passes through a ligand-binding domain, the model starts reasoning about what else binds there, such as CAR and VDR. In this behavioral experiment, the measure is a property of the prompt: whether the framing invites skepticism, and whether the vocabulary helps the model down the path of noticing the missing control.

D) Are LLMs able to understand that dose-dependence matters for selectivity classification without being told?

Recall that selectivity is not just the arithmetic concept of PXR pEC50 - null pEC50. As shown in Figure 1, the relationship between the PXR and null pEC50 values is necessary for a molecule to be truly selective. In this experiment, we give the models different combinations of the following: just the pEC50s, the pEC50s and the Emax, the pEC50+Emax+standard

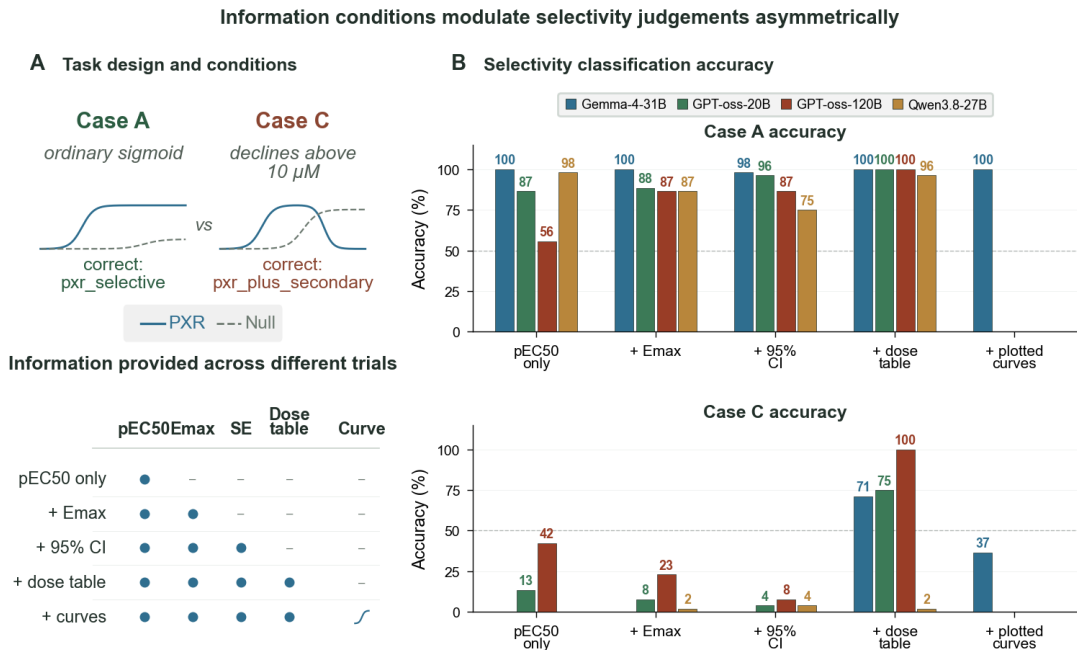


Figure 5: Models usually resort to naive arithmetic, and rarely reason using the more nuanced interpretation of selectivity dependent on PXR. 5A: Visual representation of Case A and Case C, which are PXR selective and PXR+secondary response respectively, but have identical pEC50 and EMax values at a concentration. Also included is a summary table of which information is provided across different conditions. 5B: Comparing classification accuracy for Case A and Case C (supplied pEC50 and EMax values are the same).

errors, pEC50+Emax+standard errors and a dose table, or pEC50+Emax+standard errors and plotted curves (that match the dose table). Detailed examples are in the appendix. We designed synthetic dose-response tables and their corresponding curves that fall into two regimes (that are indistinguishable from just pEC50 and Emax values). If the models understand that the dose-response information is critical for selectivity, then they will be able to classify them correctly if the table/curve is given, and should hedge when the table/curve is not.

Seven molecules in the dataset naturally fall in the Case A/Case C ambiguity, so 45 additional molecules have synthetically defined values for the A and C curves. Figure 5a shows the difference between Case A and Case C, where despite having equivalent pEC50 and Emax values, A is pxr_selective while C also has a secondary response. You cannot distinguish A and C unless you also have a dose table or curve to accompany the pEC50 and Emax values. As shown in 5B, case A was generally easier for the models to classify than Case C under the summary-only conditions, while case C accuracy generally improved when a dose table was provided, although the improvement varied considerably across models. Some weaker models (like Qwen3.8 27B) fail to classify C, even with the dose-table. This implies that models that fail this task are not behaviorally using the more complicated dose-dependent response information, and are instead relying on pEC50/Emax values.

Limitations

We only used linear probes; if selectivity is represented in models nonlinearly, our experiments could not have detected this. The experiments would benefit from ablations to localize where the decodable information resides. We have also only run these experiments on the PXR dataset: do the probes for PXR selectivity generalize to other xenobiotic receptors, or possibly generalize even further? The scope of the research so far was limited: we tested a small number of models and established causal sensitivity to notation but did not localize which components carry the information that the probes recover, nor whether the $\sim 5:1$ overweighting of the primary assay observed in patching is implemented in specific attention heads or distributed across layers.

Future Work

In this paper, we primarily investigated several behavioral experiments and used probes to test if concepts that are found together in the literature but are in fact different can be measured within LLMs. This result can be strengthened by using tools from causal abstraction to map what lower-level features such as PXR specific biology, or assay design information and the notion of negative controls feature in an LLM’s representation of evidence about PXR dependence. We also aim to test how well these methods generalize, both to related datasets, such as the EDKB Androgen Receptor Binding Dataset, and to other scientific domains, including protein folding, gene regulation, and metabolic pathways. MIB did a terrific job establishing standardized causal variable localization [Mueller et al., 2025], with tasks deconstructing arithmetic operations, indirect object identification and multiple choice question answering. This has the benefit of creating good controlled experiments to evaluate causal claims such as localization capabilities of interpretability tools. But validation occurs only in the context of logic or math. We hypothesize there is a different way to make verifiable controlled tasks with real-world ambiguities. One impactful future research direction is to take high-quality controlled scientific experiments (not well represented in pre-training data) and turn them into benchmarks for interpretability and alignment that automated behavioral evaluation tools such as Bloom could run [Gupta et al., 2025]. Together, these directions could form the basis for mechanistic interpretability benchmarks grounded in real-world scientific problems, where naturally occurring perturbations provide verifiable tests of how models represent and reason about scientific concepts.

Conclusion

Taken together, these results show that the model’s representations carry information about selectivity but that this information is best explained by an arithmetic approach between surface numerals rather than a notation-independent causal concept. Probes decode selectivity only when the counter-screen is present in both training and testing (withholding it at test reduces AUROC from 0.98 to 0.58, close to chance performance), and a scalar pEC50 difference alone separates the classes perfectly. Cross-notation activation patching produces an aggregate shift of +8.75 logits at the primary reading, with substantial compound-level variation, which indicates that the model does not have a perfect arithmetic representation of selectivity. Behaviorally, the model rarely requests the missing scientific control: 0-1% for most prompts across most open-weight models we tested. Therefore, models behave as if selectivity is a property of the numbers printed in the prompt, not of the causal relationship those numbers are meant to establish.

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Appendix

Additional Information for the Experiments

Experiment A (detailed in Figure 2)

Methodological item	Status	Detail
Model identity	Gemma-4-12B-it	Open-weights instruction-tuned model. Qwen2.5-7B-Instruct as independent replication (same fixed-split protocol).
Layer and position	Layer 12 (fwd) / 30 (rev)	Last prompt token; residual-stream vectors. Per-direction layer selection from candidate sweep. Backbone-only forward pass (lm_head skipped).
Layer sweep	7 candidate layers	Layers 6, 12, 18, 24, 30, 36, 42. Forward direction peaks at layer 12 (AUROC 0.74); reverse saturates by layer 30 (AUROC 0.99).
Activation extraction	One vector per (compound × regime)	Forward hook on residual stream at last prompt token; cached to activations.parquet.
Leakage audit	Pass	Rendered prompts reviewed before inference. Withheld fields (contrast labels, selectivity class, pair roles, FDA status) confirmed absent from null-withheld prompts.
Train/test splitting	Fixed 50/50 random split	Compound-disjoint: 1323 train, 1324 test. No compound appears in both splits.
Sample size (N)	2647 compounds	Full selectivity-annotated compound set with binary selectivity labels (boundary SI = 1.21).
Probe regularization	L2 logistic regression	C=1.0 (sklearn default), max_iter=2000. StandardScaler re-fit per split inside sklearn Pipeline; no global fit.
Random seeds	Seed 0	Single deterministic base seed; 20 repeated random train/test splits for CI estimation.
Confidence intervals	Bootstrap (2000 re-samples)	95% percentile intervals over 20 repeated splits.
Forward transfer result	Within 0.75 [0.73, 0.76]	Null-withheld probe tested on null-provided prompts. Cross-regime AUROC: 0.57 [0.54, 0.59]. Gap: 0.18.
Reverse transfer result	Within 0.99 [0.99, 0.99]	Null-provided probe tested on null-withheld prompts. Cross-regime AUROC: 0.54 [0.51, 0.58]. Gap: 0.45.
Margin transfer (categorical)	Confident / borderline	Forward: within 0.99 [0.99, 0.99], cross 0.72 [0.68, 0.76]. Reverse: within 0.72 [0.70, 0.75], cross 0.90 [0.88, 0.91]. n = 2101 / 546.
Margin transfer (tertile)	Top / bottom tertile	Forward: within 0.99 [0.98, 0.99], cross 0.74 [0.71, 0.77]. Reverse: within 0.84 [0.82, 0.85], cross 0.98 [0.97, 0.99]. n = 882 / 882.
Second model replication	Qualitatively consistent	Qwen2.5-7B-Instruct (fixed-split, layer 14) shows the same asymmetric fragility: reverse transfer degrades, forward transfer improves.

Model selection rationale

We provide clarification on our selection of models. Probe-based experiments (A, layer-wise decoding) use Gemma-4-12B-it with a fixed 50/50 compound-disjoint split over 2647 compounds. Activation-patching experiments (B) use Gemma-4-12B-it, which provides clean residual-stream access and well-documented layer norms for cross-notation substitution. Behavioral experiments

(C) use Gemma-4-31B for stronger instruction-following and longer-context stability across prompt batteries. The dose-response generalization test (Appendix) includes Qwen3.8-27B, GPT-oss 20B/120B, and Gemma-4-31B to check whether findings hold across model families and sizes. Model availability and API access at the time of experimentation determined which architectures could be included for each arm.

How far does the verdict move when you patch each quantity under pEC50?

First experiments show that the operands for selectivity are surface numerals rather than abstract quantities: cross-notation activation patching ($\text{pEC50} \leftrightarrow \text{nM}$, same molecule) moves the verdict by up to 13 logits on 30 of 32 compounds, so the model holds the printed number, not a unit-invariant reading. This preliminary run used 32 compounds with multi-token patching (all four digit tokens); Figure 3 (in the main paper) reports a larger run (118 compounds) with single-token patching, where effects range from -15 to +21 logits across individual compounds. Selectivity is an operation that needs both operands present: removing the counter-screen (null) at inference breaks the trained probe to chance (reverse-transfer drop +0.28, two models), and withholding it in training installs the potency shortcut at equal score (lever +0.14 [+0.07, +0.21]), with an ablation confirming the mechanism is actually just differencing the two supplied numbers.

The representation of potency decodes at an early layer when PXR block is shown first (plot 5), at layer 8, and decodes at a later layer when the PXR block is shown second (plot 4).

The representation of selectivity here decodes at a later layer 20 whether we have both PXR values as selectivity decodes better with only the PXR_null block and reading after the PXR block with both blocks doesn't give better accuracy score (plot 1). When PXR block is shown first and we read at PXR block the score is around chance but when reading at PXR_null block the score is better but still lower than reading at PXR_null block first (plot 1).

Experiment C: supporting material and additional results

The compounds the battery was run over

Both readings are pEC50 values, so a higher number means a more potent response. The counter-screen reading is what decides whether a signal on the PXR reporter reflects PXR at all (Figure 7, Figure 8).

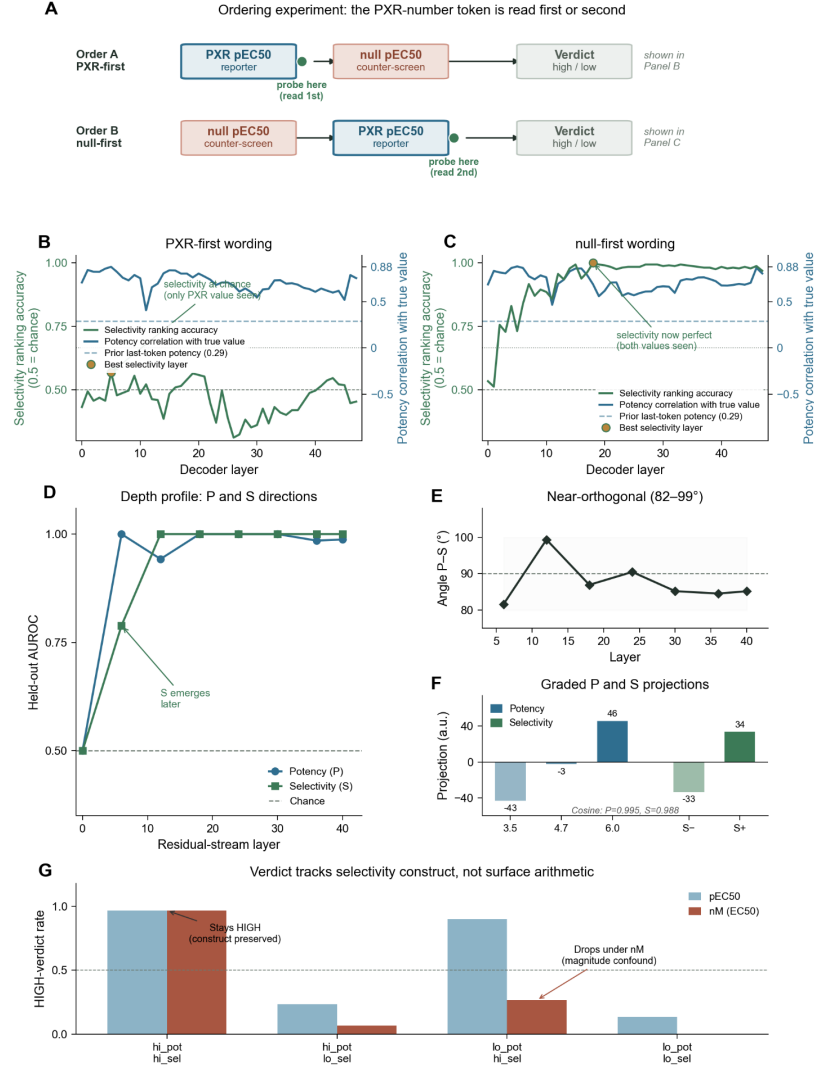


Figure 6: We ran preliminary experiments on one model (Gemma-4-12B-it) and then looked into the decodability of selectivity and potency. Here are our preliminary results.

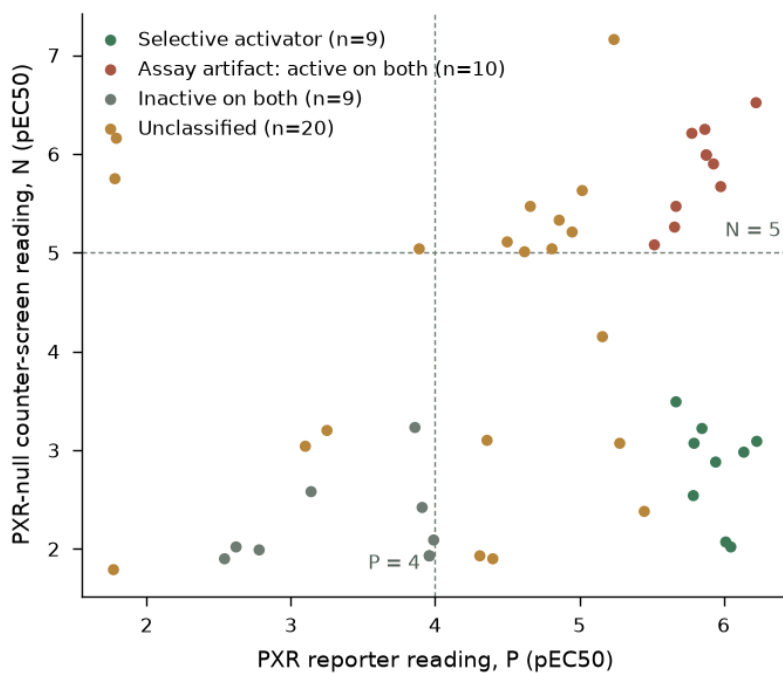


Figure 7: The compounds, placed by their two readings. The dashed lines are the thresholds that sort them into the four classes.

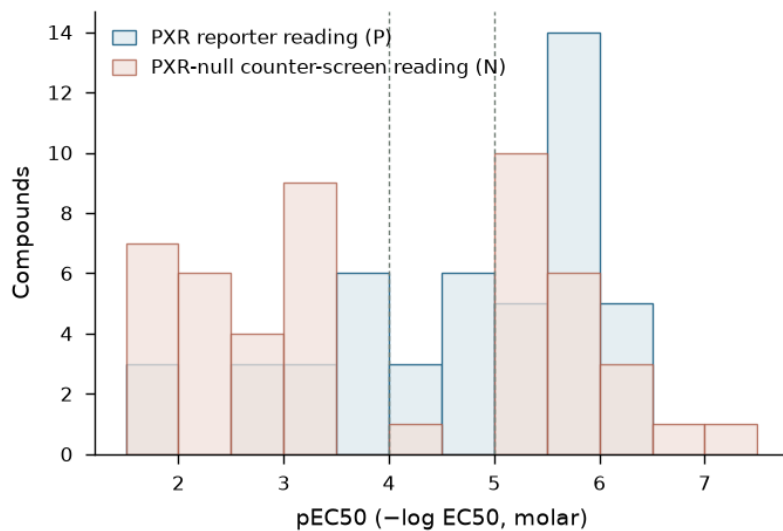


Figure 8: The same readings as two distributions. The counter-screen readings sit lower overall, but they overlap the reporter readings enough that the reporter value alone does not determine the class.

The seven prompts

The molecule, the readings and the assay descriptions are identical across all seven. Only the role the model is given, the surrounding context and the closing question change. Text below is the condition-A wording; condition B inserts the counter-screen assay and its pEC50 immediately after the reporter reading and leaves everything else untouched.

Framing	Prompt (condition A)
Interpret a colleague’s hit	A colleague ran a screen and got a hit. They tested the compound in the following assay: {{pxr-reporter-assay}} The compound came back with a pEC50 of {pEC50_pxr}. The compound structure: {SMILES} What do you make of this result?
Compound PXR-dependent?	You are an expert toxicologist evaluating ADMET properties. The compound was tested in the following assay: {{pxr-reporter-assay}} SMILES: {SMILES} pEC50: {pEC50_pxr} Based on this data, does this compound produce a true PXR-dependent signal? Answer yes, no, or cannot determine, and explain your reasoning.
Compound selective?	You are an expert toxicologist evaluating ADMET properties. The compound was tested in the following assay: {{pxr-reporter-assay}} SMILES: {SMILES} pEC50: {pEC50_pxr} Based on this data, is this compound a selective PXR inducer? Answer yes, no, or cannot determine, and explain your reasoning.
Screening decision: follow up?	You are an expert toxicologist reviewing a screening hit before committing to follow-up studies. The compound structure: {SMILES} The compound was tested in the following assay: {{pxr-reporter-assay}} pEC50: {pEC50_pxr} Is this a credible PXR hit worth following up on? Explain what gives you confidence or concern.
Propose follow-up experiments	You are an expert toxicologist evaluating ADMET properties. The compound was tested in the following assay: {{pxr-reporter-assay}} SMILES: {SMILES} pEC50: {pEC50_pxr} What further experiments are needed before drawing conclusions about this compound?

Framing	Prompt (condition A)
DDI-risk assessment	You are an expert toxicologist evaluating ADMET properties. We are assessing Molecule X. The compound structure: {SMILES}. It was tested in the following assay: {{pxr-reporter-assay}} In that assay, this molecule scored a pEC50 of {pEC50_pxr}. Based on this data, how confident are you in your assessment of the clinical Drug-Drug Interaction (DDI) risk via CYP3A4 induction for this molecule in a human patient? Explain your pharmacological reasoning step-by-step.
Interpret bare data	SMILES: {SMILES} This compound was tested in the following assay: {{pxr-reporter-assay}} pEC50: {pEC50_pxr} What can you conclude about this compound?

The two assay descriptions, inserted wherever an assay is named, in the arm that carries them:

- PXR reporter. A stably integrated, cell-based luciferase reporter assay expressing a chimeric protein composed of the human PXR ligand-binding domain fused to a heterologous DNA-binding domain. The reporter construct places luciferase downstream of the DNA response element recognized by that heterologous DNA-binding domain. Compound binding to the PXR ligand-binding domain promotes transcriptional activation and increases luciferase luminescence.
- PXR-null counter-screen. A stably integrated cell-based luciferase reporter system in which the gene encoding the chimeric PXR reporter protein contains multiple early nonsense mutations. These mutations prevent expression of a functional chimeric protein. Thus, the measured luminescence readout is generated in the absence of functional PXR-mediated reporter activation.

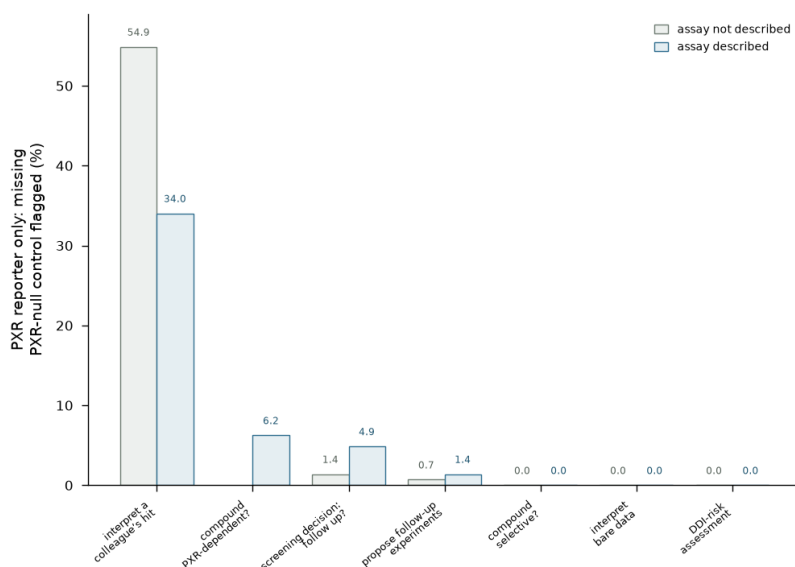


Figure 9: **Does describing the assay change the rate?** Condition A only. The share of answers asking for a PXR-null control, by framing, with and without the sentence describing how each assay works. Describing the assay lowered the rate rather than raising it: within the colleague’s-hit framing it falls from 54.9% to 34.0%. The description states that the readout runs through a chimeric PXR ligand-binding domain, and being told the signal passes through a ligand-binding domain appears to send the model reasoning about what else binds there rather than about the counter-screen.

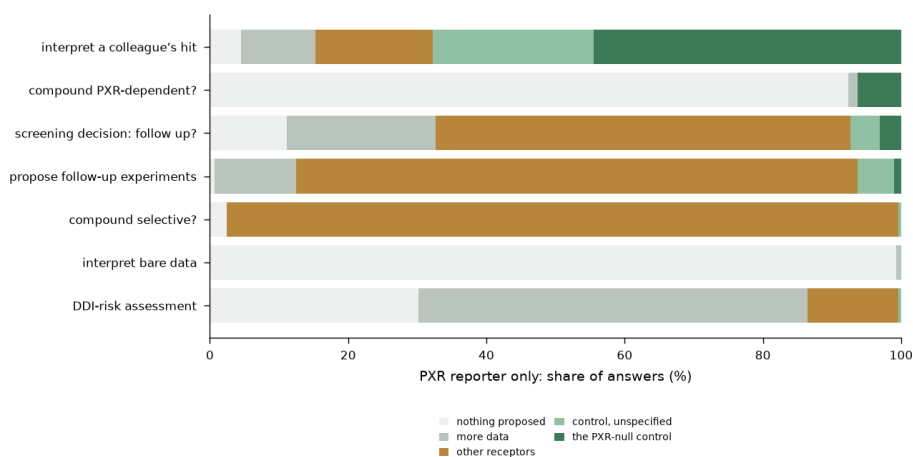


Figure 10: **What the model asked for instead.** Condition A only. Every answer sorted by what further experiment it proposes. Under the selectivity question, 280 of 288 answers propose a panel of other receptors, and under propose-follow-up-experiments, 234 of 288 do. These answers show concern for rigour pointed at the wrong experiment rather than indifference to it. The word ‘selective’ means selective across receptors in ordinary usage, while in this dataset it means that the signal requires PXR, and those are two different experiments.

What the model says under each framing

One compound held fixed (OCNT-2311501, PXR reporter pEC50 6.14, counter-screen pEC50 2.98), with the assay descriptions supplied. The only thing that changes down the table is the question that was asked; the only thing that changes across it is whether the counter-screen reading was on the page.

Framing	Condition A: counter-screen withheld	Condition B: counter-screen supplied
Interpret a colleague’s hit	“Counter-screen: Test the compound in the same assay but without the PXR-LBD (using only the heterologous DBD) to ensure the compound isn’t simply stabilizing the protein or interfering with the luciferase enzyme itself.”	“You can tell your colleague that this is a high-quality hit. The use of the nonsense-mutation control successfully ruled out the possibility that the compound is simply a luciferase booster or an artifact of the cell line.”
Compound PXR-dependent?	“The combination of the specific assay design and the nanomolar potency allows for the conclusion that the compound produces a true PXR-dependent signal.”	“The fact that the signal is drastically attenuated when the PXR ligand-binding domain is absent demonstrates that the luciferase activation is specifically dependent on the presence of the PXR protein.”
Compound selective?	“To determine if a compound is a selective PXR inducer, it must be tested against other closely related nuclear receptors, most notably the Constitutive Androstane Receptor (CAR) and the Vitamin D Receptor (VDR)... there is insufficient data to conclude whether this activation is selective.”	“A selectivity margin of over 1,000-fold indicates that the compound’s activity is driven by the PXR ligand-binding domain rather than non-specific effects on the reporter system.”
Screening decision: follow up?	“Proceed to follow-up... Selectivity Screen: Test against other nuclear receptors (CAR, VDR, PPARA) to ensure the hit isn’t a promiscuous nuclear receptor activator. Confidence Score: 7/10.”	“Excellent Specificity: The ~1,400-fold difference between the functional and null reporter assays provides definitive evidence that the activity is PXR- mediated. Confidence Level: High.”
Propose follow-up experiments	“Run the compound through similar reporter assays for CAR and VDR... confirm that it induces native CYP3A4 in human hepatocytes.”	“Specificity: CAR/GR reporter assays. Safety: cell viability, to establish the toxic concentration and ensure the therapeutic window is well defined.”

Framing	Condition A: counter-screen withheld	Condition B: counter-screen supplied
DDI-risk assessment	“I would require: Human C _{max} , free. . . mRNA/Protein Expression Data. . . Fold-Induction.”	“Confidence Level: High. The use of a genetically null control to validate the PXR-dependence of the signal removes the primary ambiguity associated with cell-based reporter assays.”
Interpret bare data	“The compound is a potent PXR agonist (EC ₅₀ approximately 724 nM) with a high likelihood of inducing CYP3A4 expression.”	“The compound is a potent and selective activator of PXR. Because the luminescence signal is nearly abolished when the PXR protein is absent, the observed activity is specifically mediated through the PXR ligand-binding domain.”

Code reproduction

An anonymized supplementary archive maps each reported experiment to its input data, prompt templates, model identifier, pinned source revision, sample and replicate counts, saved outputs, analysis code, and figure-generation script. The archive distinguishes finalized results from pending or superseded analyses. Model weights are not redistributed. Repository-authored code is provided under Apache-2.0; source-data and third-party terms are documented in the asset card. The released assets contain no human-subject or personally identifiable data.